

Driven Motion

Suppose that an external force $F_e(t)$ is applied to the mass-spring system.

$$\vec{F} = m \vec{a}$$

$$\begin{array}{ccccccc} mg & - & k(s+x(t)) & - & \beta x'(t) & + & F_e(t) = m x''(t) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \text{weight} & & \text{spring force} & & \text{damping Force} & & \text{external Force} & & ma \end{array}$$

Since $mg - ks = 0$, then:

$$m x''(t) + \beta x'(t) + k x(t) = F_e(t)$$

ex: ④ A mass of 1 kg is attached to a spring of constant 3 N/m. There is a damping force of magnitude four times the instantaneous velocity of the mass. An external force (in N) of the form:

$$F_e(t) = 2 \sin t$$

is applied to the system. Find the position of the mass if it is initially at rest in the equilibrium position.

$x(t)$ = position of the mass at time t

$$m x''(t) + \beta x'(t) + k x(t) = F_e(t)$$

$$\begin{cases} m = 1 \text{ kg} \\ \beta = 4 \frac{\text{N} \cdot \text{s}}{\text{m}} \\ k = 3 \text{ N/m} \end{cases}$$

$$\text{and } F_e(t) = 2 \sin t \text{ (N)}$$

$$\Rightarrow x''(t) + 4x'(t) + 3x(t) = 2 \sin t$$

The initial conditions are

$$\begin{cases} x(0) = 0 & (\text{from equilibrium}) \\ x'(0) = 0 & (\text{from rest}) \end{cases}$$

Step ① Find $x_c(t)$

Roots : $\lambda^2 + 4\lambda + 3 = 0$
 $(\lambda + 3)(\lambda + 1) = 0$
 $\lambda = -3, -1$

$$\Rightarrow x_c(t) = c_1 e^{-3t} + c_2 e^{-t}$$

Step ② Find $x_p(t)$

Since the RHS of the DE is $2 \sin t$,
we look for x_p as :

$$x_p = a \sin t + b \cos t$$

$$x'_p = a \cos t - b \sin t$$

$$x''_p = -a \sin t - b \cos t$$

Plug x_p , x'_p , and x''_p in the DE to get:

$$\begin{aligned} & [-a \sin t - b \cos t] + 4[a \cos t - b \sin t] + 3[a \sin t + b \cos t] \\ & \dots = 2 \sin t \end{aligned}$$

$$\begin{aligned} \Rightarrow & (-a - 4b + 3a) \sin t + (-b + 4a + 3b) \cos t \\ & \dots = 2 \sin t + 0 \cos t \end{aligned}$$

$$\Rightarrow \begin{cases} 2a - 4b = 2 \\ 4a + 2b = 0 \end{cases}$$

$$\text{solve to get : } a = \frac{1}{5} \text{ and } b = -\frac{2}{5}.$$

The general solution is :

$$x(t) = x_c(t) + x_p(t)$$

$$x(t) = c_1 e^{-3t} + c_2 e^{-t} + \frac{1}{5} \sin t - \frac{2}{5} \cos t$$

Step ③ Find c_1, c_2

$$x'(t) = -3c_1 e^{-3t} - c_2 e^{-t} + \frac{1}{5} \cos t + \frac{2}{5} \sin t$$

$$x(0) = 0 \rightarrow c_1 + c_2 - \frac{2}{5} = 0$$

$$x'(0) = 0 \rightarrow -3c_1 - c_2 + \frac{1}{5} = 0$$

$$\begin{cases} c_1 + c_2 = 2/5 \\ 3c_1 + c_2 = 1/5 \end{cases} \xrightarrow{\text{solve}} \begin{cases} c_1 = -1/10 \\ c_2 = 1/2 \end{cases}$$

The position of the mass is then:

$$x(t) = -\frac{1}{10} e^{-3t} + \frac{1}{2} e^{-t} + \frac{1}{5} \sin t - \frac{2}{5} \cos t$$

Since $\lim_{t \rightarrow \infty} x_c(t) = 0$,

$x_c(t)$ is a Transient.

$x_p(t) = \frac{1}{5} \sin t - \frac{2}{5} \cos t$ is called

the steady-state solution.

ex: ⑤

Beats and Resonance

Consider a mass m attached to a spring of constant K with negligible damping ($\beta \approx 0$) and driven by an external

force $F_e(t) = F_0 \cos(\gamma t)$.

Find the position of the mass $x(t)$ if it is initially at rest in the equilibrium position.

We have to solve the IVP

$$\begin{cases} m x''(t) + K x(t) = F_0 \cos(\gamma t) \\ x(0) = 0 & \text{(equilibrium)} \\ x'(0) = 0 & \text{(rest)} \end{cases}$$

By dividing by m , the DE becomes

$$x''(t) + \omega^2 x(t) = \frac{F_0}{m} \cos(\gamma t)$$

where $\omega^2 = \frac{k}{m}$.

We will consider two cases for γ .

Case ① $\gamma \neq \omega$

Case ② $\gamma = \omega$

Assume $\gamma \neq \omega$

Let's solve the IVP.

step ① Find x_c

$$x_c = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

Step (a) Find x_p

Let's look for x_p as

$$x_p = a \cos(\gamma t) + b \sin(\gamma t)$$

[No bad case since $\gamma \neq \omega$].

$$\Rightarrow x_p' = -a\gamma \sin(\gamma t) + b\gamma \cos(\gamma t)$$

$$\Rightarrow x_p'' = -a\gamma^2 \cos(\gamma t) - b\gamma^2 \sin(\gamma t)$$

Plug x_p , x_p' , and x_p'' in the DE to get:

$$x_p'' + \omega^2 x_p = \frac{F_0}{m} \cos(\gamma t)$$

$$\left[-a\gamma^2 \cos(\gamma t) - b\gamma^2 \sin(\gamma t) \right] + \omega^2 \left[a \cos(\gamma t) + b \sin(\gamma t) \right]$$

$$\dots = \frac{F_0}{m} \cos(\gamma t)$$

$$\Rightarrow (-a\gamma^2 + a\omega^2) \cos(\gamma t) + (-b\gamma^2 + b\omega^2) \sin(\gamma t) \\ \dots = \frac{F_0}{m} \cos(\gamma t) + 0 \sin(\gamma t)$$

$$\Rightarrow \begin{cases} -a\gamma^2 + a\omega^2 = \frac{F_0}{m} \\ -b\gamma^2 + b\omega^2 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} a = \frac{F_0}{m(\omega^2 - \gamma^2)} \\ b = 0 \end{cases}$$

The general solution is then:

$$x(t) = x_c(t) + x_p(t)$$

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t) + \frac{F_0}{m(\omega^2 - \gamma^2)} \cos(\gamma t)$$

if $\gamma \neq \omega$

Step ③ Find c_1, c_2

$$x'(t) = -c_1 \omega \sin(\omega t) + c_2 \omega \cos(\omega t) - \frac{F_0 \gamma}{m(\omega^2 - \gamma^2)} \sin(\gamma t)$$

$$x(0) = 0 \rightarrow c_1 + 0 + \frac{F_0}{m(\omega^2 - \gamma^2)} = 0$$

$$x'(0) = 0 \rightarrow 0 + c_2 \omega - 0 = 0$$

$$\Rightarrow \begin{cases} c_1 = -\frac{F_0}{m(\omega^2 - \gamma^2)} \\ c_2 = 0 \end{cases}$$

If $\gamma \neq \omega$, the IVP solution is

$$x(t) = \frac{F_0}{m(\omega^2 - \gamma^2)} \left[\cos(\gamma t) - \cos(\omega t) \right]$$

By using the trig identity :

$$\cos A - \cos B = 2 \sin\left(\frac{B-A}{2}\right) \sin\left(\frac{B+A}{2}\right)$$

With $A = \gamma t$ and $B = \omega t$, the IVP solution becomes the following.

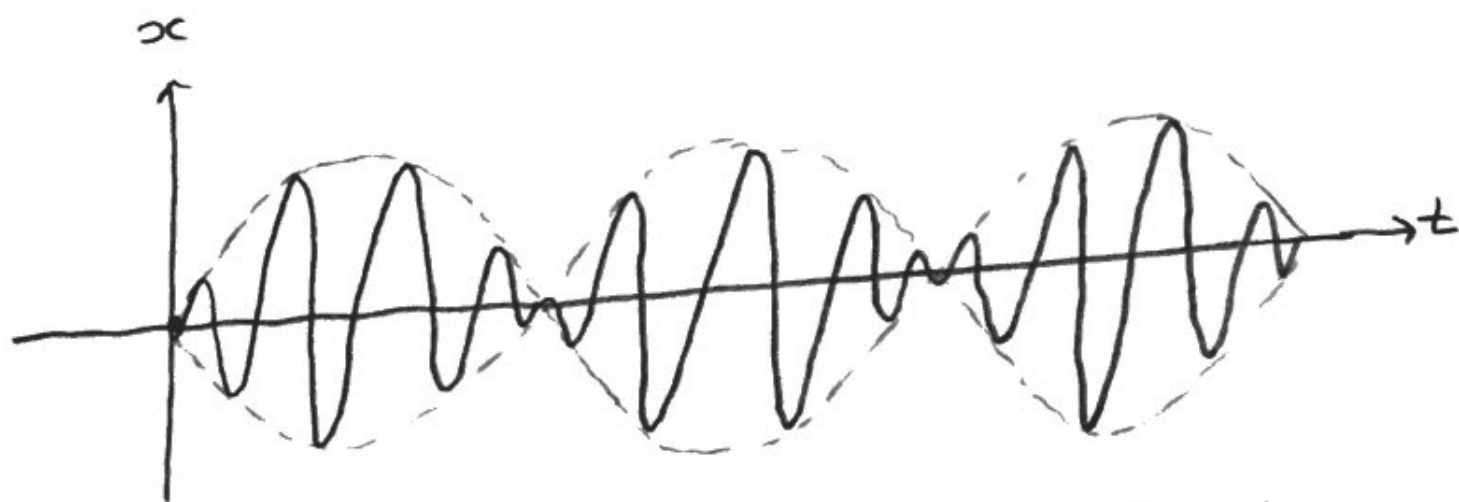
$$x(t) = \frac{2F_0}{m(\omega^2 - \gamma^2)} \sin\left(\left(\frac{\omega - \gamma}{2}\right)t\right) \sin\left(\left(\frac{\omega + \gamma}{2}\right)t\right)$$

To understand the behavior of $x(t)$

$$x(t) = \left[\frac{a F_0}{m(\gamma^2 - \omega^2)} \sin\left(\left(\frac{\omega - \gamma}{2}\right)t\right) \right] \sin\left(\left(\frac{\omega + \gamma}{2}\right)t\right)$$

↓
"Amplitude"

We can think of the lower frequency portion as the "Amplitude" of the higher frequency term. A typical graph looks like this.



Oscillations like these are called beats.

Let's consider the case $\gamma = \omega$.

The IVP for this case is

$$\begin{cases} x''(t) + \omega^2 x(t) = \frac{F_0}{m} \cos(\omega t) \\ x(0) = 0 \\ x'(0) = 0 \end{cases}$$

Instead of solving this IVP directly,
we can obtain its solution by
taking the limit as $\gamma \rightarrow \omega$
of the solution when $\gamma \neq \omega$.

If $\gamma \neq \omega$, the solution was:

$$x(t) = \frac{F_0}{m(\omega^2 - \gamma^2)} [\cos(\gamma t) - \cos(\omega t)]$$

If $\gamma = \omega$, the IVP solution is

$$x(t) = \lim_{\gamma \rightarrow \omega} \frac{F_0 [\cos(\gamma t) - \cos(\omega t)]}{m(\omega^2 - \gamma^2)}$$

we get a $\frac{0}{0}$ limit so let's use L'Hôpital. The variable is γ , everything else is kept constant.

$$\Rightarrow x(t) \stackrel{(H)}{=} \lim_{\gamma \rightarrow \omega} \frac{F_0 (-t \sin(\gamma t))}{m(-2\gamma)}$$

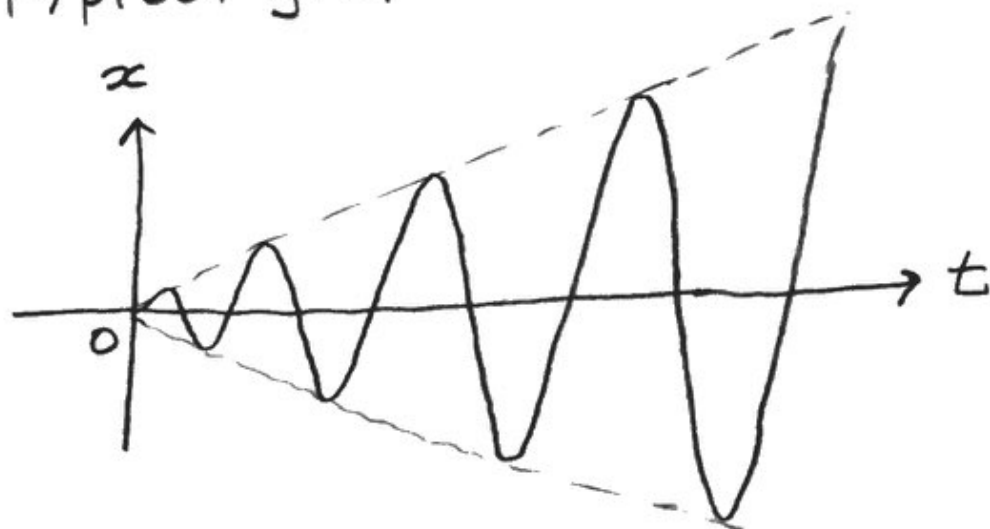
$$\Rightarrow x(t) = \frac{F_0 t}{2m\omega} \sin(\omega t)$$

To understand how $x(t)$ behaves,
think of $\frac{F_0 t}{2m\omega}$ as the "Amplitude"
of $\sin(\omega t)$.

$$x(t) = \left[\frac{F_0 t}{2m\omega} \right] \sin(\omega t)$$

↓
"Amplitude"

A typical graph looks like this.



This is an example of Resonance.
An oscillation with an Amplitude that
grows with time.